

Anthony Liang

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RESEARCH INTERESTS

My research focuses on scalable robot learning, with an emphasis on data-efficient human feedback (gaze, demonstrations, preferences, interventions) and reward foundation models that provide generalizable reward functions across tasks, embodiments, and environments. I also work on core components for embodied intelligence, including vision-language-action models for low-level control, trace models for structured planning, and world models for scene understanding and predictive reasoning. In parallel, I am investigating RL post-training for LLM-based web agents to improve reliability in sequential UI tasks and long-horizon interactions.

EDUCATION

University of Southern California , Los Angeles, CA <i>Ph.D.</i> in Computer Science (Co-advised by Erdem Biyik and Stephen Tu)	Aug 2021 - May 2026 <i>GPA: 3.9/4.0</i>
University of Michigan Rackham Graduate School , Ann Arbor, MI <i>M.S.</i> in Robotics (Advisor: Honglak Lee)	Aug 2017 - 2021 <i>GPA: 4.0/4.0</i>
University of Michigan , Ann Arbor, MI <i>B.S.E.</i> (Advisor: Honglak Lee)	Aug 2017 - 2021 <i>GPA: 3.65/4.0</i>

RESEARCH EXPERIENCE

LiRA Lab and Statistical Learning Lab , USC <i>Ph.D. Student, PI: Erdem Biyik, Stephen Tu</i> Robot learning with multimodal human feedback, sample-efficient RL, learning from unlabelled videos	Aug 2021 - May 2026
Intelligent Robot Lab , Carnegie Mellon <i>Visiting Researcher</i> with Changliu Liu Hierarchical RL for safe control of autonomous vehicles in dynamic environments	May 2020 - May 2021
Deep Learning Lab , University of Michigan <i>Research Intern</i> with Honglak Lee Sample-efficient RL for embodied task learning	Jan 2019 - May 2021

PROFESSIONAL EXPERIENCE

Toyota Research Institute <i>Research Scientist</i> Post-training and sample-efficient real-world RL, policy improvement in World-Action Models, learning from suboptimal data	May 2026 - Present <i>Los Altos, CA</i>
Google DeepMind <i>Research Intern</i> with Kalpesh Krishna, Jacob Eisenstein RL post-training for reasoning models	May 2025 - Sept 2025 <i>NYC/Seattle</i>
Google Research <i>Research Intern</i> with Chih-Wei Hsu, Yinlam Chow, Guy Tennenholtz, Craig Boutilier - Bayesian RL for Markov Decision Processes with gradually changing latent dynamics - Data augmentation at critical states for robot imitation learning with Stephen Tu - Generative modeling for online RL policies	May 2023 - Feb 2024 <i>Remote NYC</i>
Meta AI - Multimodal Learning Team <i>Research Intern</i> with Paul Crook and Andrea Madatto Fine-tuning large language models for task-oriented dialogue generation	May 2022 - Aug 2022 <i>Redmond, WA</i>
Amazon Science <i>Applied Science Intern</i> with Thiago Mosquero Collaborative filtering for recommending new brands and products to consumers	May 2021 - Aug 2021 <i>Seattle, WA</i>

Invisible.ai

AI Research Intern

May 2020 - Aug 2020

Remote

- Improving computer vision models for real-time object detection and tracking for industrial processes

Google Ads

Software Engineering Intern

May 2019 - Aug 2019

Mountain View, CA

Luminar Technologies

Software Engineering Intern

May 2018 - Aug 2018

Palo Alto, CA

PUBLICATIONS

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- [C7] **Anthony Liang***, Yigit Korkmaz*, Jiahui Zhang, Minyoung Hwang, Abrar Anwar, Sid Kaushik, Aditya Shah, Alex Huang, Luke Zettlemoyer, Dieter Fox, Yu Xiang, Anqi Li, Andreea Bobu, Abhishek Gupta, Stephen Tu[†], Erdem Biyik[†], Jesse Zhang[†]. "Robometer: Scaling General-Purpose Robotic Reward Models via Trajectory Comparisons". *Robotics: Science and Systems (RSS) 2026*
- [C6] **Anthony Liang**, Jonathan Berant, Adam Fisch, Abhimanyu Goyal, Kalpesh Krishna[†], Jacob Eisenstein[†]. "Plantain: Plan-Answer Interleaved Reasoning". *International Conference on Machine Learning (ICML) 2026 (Spotlight)*
- [C5] Matthew Hong*, **Anthony Liang***, Kevin Kim, Harshitha Rajaprakash, Jesse Thomason[†], Erdem Biyik[†], Jesse Zhang[†]. "HAND Me the Data: Fast Robot Adaptation via Hand Path Retrieval" *International Conference on Robot Automation (ICRA) 2026*
- [C4] **Anthony Liang***, Pavel Czempin*, Matthew Hong, Yutai Zhou, Erdem Biyik, Stephen Tu. "CLAM: Continuous Latent Action Models for Robot Learning from Unlabeled Demonstrations" *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) 2026*
- [C3] **Anthony Liang**, Chih-wei Hsu, Yinlam Chow, Guy Tennenholtz, Erdem Biyik, Craig Boutilier. "DynaMITE-RL: A Dynamics Model for Improved Temporal Meta Reinforcement Learning", *International Conference on Machine Learning (ICML) AutoRL Workshop 2024, Conference on Neural Information Processing Systems (NeurIPS) 2024*
- [C2] **Anthony Liang**, Jesse Thomason, Erdem Biyik. "ViSaRL: Visual Reinforcement Learning Guided By Human Saliency", *Spotlight talk at ICRA Pretraining for Robotics (PT4R) Workshop, IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) 2024*
- [C1] Wilka Carvalho, **Anthony Liang**, Kimin Lee, Sungryull Sohn, Honglak Lee, Richard L. Lewis, Satinder Singh. "Reinforcement Learning for Sparse-Reward Object-Interaction Tasks in a First-person Simulated 3D Environment", *International Joint Conferences on Artificial Intelligence (IJCAI) 2021*
- [T5] **Anthony Liang**, Yigit Korkmaz, Jiahui Zhang, Jesse Zhang, Abrar Anwar, Sidhant Kaushik, Yufei Wang, Yu Xiang, David Held, Dieter Fox, Abhishek Gupta, Stephen Tu, Erdem Biyik. "SPUR: Scaling Reward Learning from Human Demonstrations". *CoRL Eval&Deploy Workshop 2025*
- [T4] **Anthony Liang**, Pavel Czempin, Yutai Zhou, Stephen Tu, Erdem Biyik. "In-Context Generalization to New Tasks From Unlabeled Observation Data", *ICML In-Context Learning Workshop 2024*
- [T3] Ishika Singh, **Anthony Liang**, Mohit Shridhar, Jesse Thomason. "Self-Supervised 3D Representation Learning for Robotics", *ICRA Pretraining for Robotics (PT4R) 2023*
- [T2] **Anthony Liang**, Ishika Singh, Karl Pertsch, Jesse Thomason. "Transformer Adapters for Robot Learning", *CoRL Workshop on Pretraining for Robot Learning 2022*
- [T1] Wilka Carvalho, **Anthony Liang**, Kimin Lee, Sungryull Sohn, Richard L. Lewis, Satinder Singh, Honglak Lee. "ROMA: A Relational, Object-Model Learning Agent for Sample-Efficient Reinforcement Learning", *ICML Workshop on Object-Oriented Learning 2020*

TEACHING

University of Southern California

CSCI 445: Foundation of Robotics

Fall 2026

CSCI 699: Robot Learning

Fall 2024

CSCI 499: Natural Language for Interactive AI

Fall 2022

University of Michigan, Ann Arbor

EECS 442: Computer Vision

Winter 2021

EECS 498: Algorithmic Robotics

Fall 2020

EECS 504: Graduate Computer Vision

Winter 2020

EECS 280: Introduction to Programming and Data Structures

Fall 2018 - Fall 2019

Summer STEM Institute Research Mentor

Summer 2021

SERVICES

Reviewer:

- ICML, ICLR, ICRA, NeurIPS, HRI, RA-L, AAAI, RO-MAN, CVPR

Workshop Organizer:

- Structured Physical Intelligence (SPIN) at CoRL 2026
- Human-in-the-Loop Robot Learning: Teaching, Correcting, and Adapting at RSS 2025

Mentorship and Outreach:

- USC UGrad Mentoring Program
- CURVE: Mentored USC undergrads through fellowship program.

STUDENT MENTORSHIP

USC Masters Students

- Matthew Hong Learning from unlabelled data, RLHF
- Dhanush Kumar Penmetsa Gaze for robot teleoperation
- Ziyi Liu Preference-based RL

USC Undergraduate Students

- Yixi Quan (Undergrad, USC) Pretrained Video-LLMs for Robot Learning
- Jaiv Doshi (Undergrad, USC) Human-intervention reinforcement learning
- Junu Song (USC CURVE Fellowship) Real-world robot navigation

Fellowship Programs and External Collaborators

- Mihir Rao (Undergrad, Berkeley) Trace VLAs
- Sankalp (Sunny) Agrawal (Undergrad, USC SURE Program) Meta-RL with task descriptors
- Shreya Ramanujam (Undergrad, IIT) Gaze for robot teleoperation

INVITED TALKS

"Scaling Reward Models via Preference Comparisons"

2026

- GRAIL Lab, NYU (Host: Lerrel Pinto)
- Apple Robotics Reading Group (Host: Soheil Habibiian)

HONORS AND AWARDS

- NSF Graduate Research Fellowship Honorable Mention 2020